Cyber Security Analys

Threat Detection with Splunk
Dataset: Kaggle Cyber Attacks

Name: Lucas Camargo

Dataset Overview

Source: Kaggle – "Cyber Security Attacks" (Team Incribo).

Volume: 40,000 synthetic but **realistic cyber attack events**.

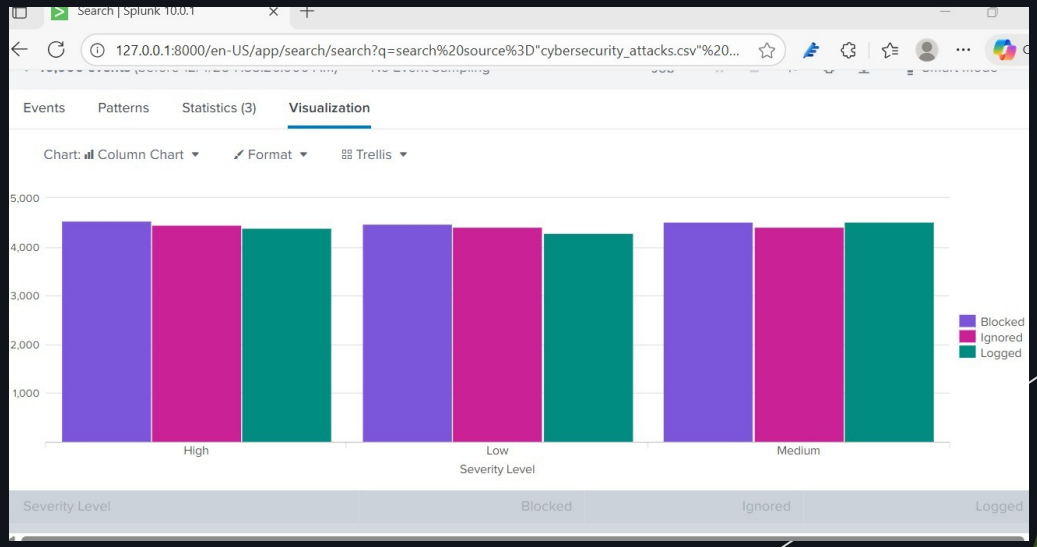
Key fields: Timestamp, Source/Destination IP, Attack Type, Severity Level, Anomaly Score.

Why this dataset: Large enough **to simulate a Security Operations Centre (SOC) log environment.**

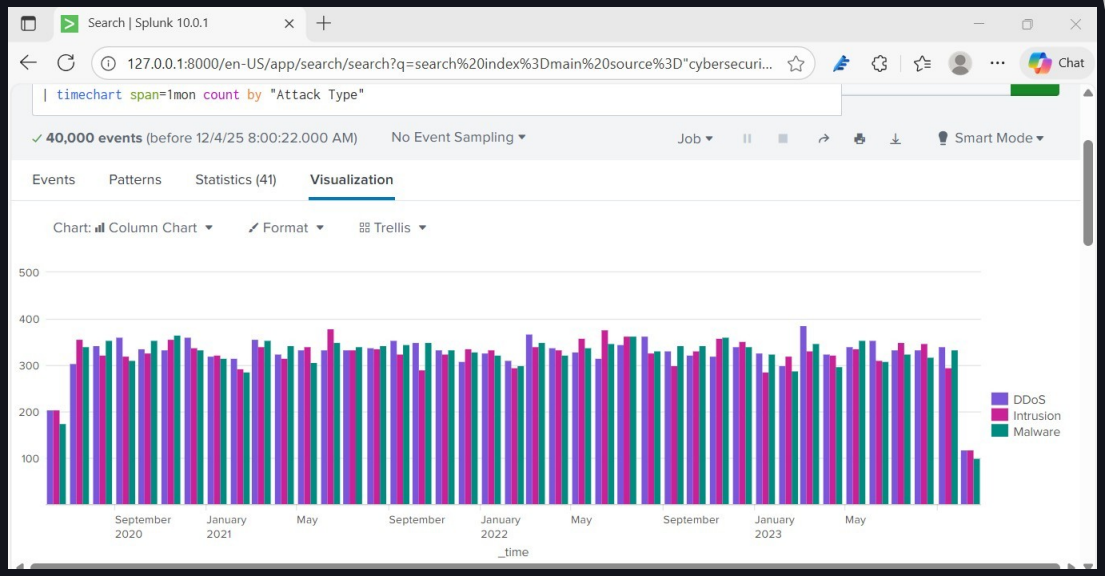
Use in Splunk: Ideal for practising threat hunting, discrepancy checks, trends, and anomaly

Discrepancy: Policy Fail

This chart shows how the **firewall responded to different severities**. A major issue stands out: **many <u>High-Severity attacks were ignored instead of blocked</u>,** which indicates a policy or configuration failure.



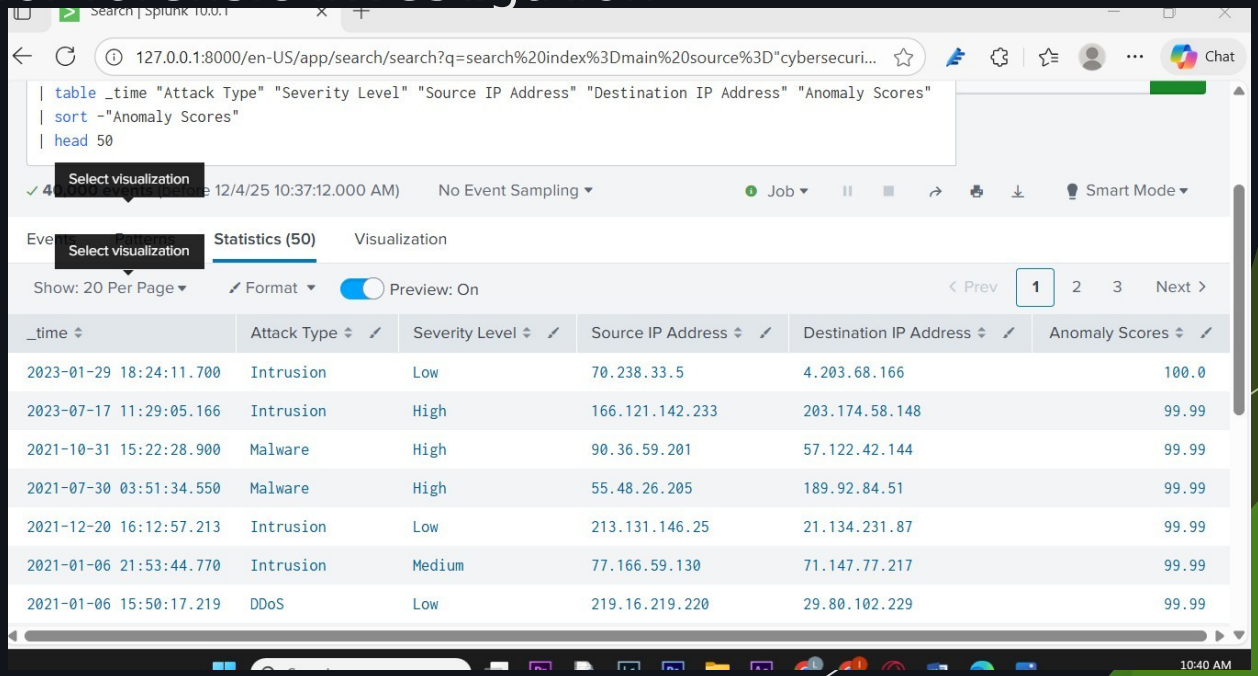
Attack Trends Over Tim



DDoS, Intrusion, and **Malware events** appear every single month at **stable levels**. This tells us the organization is under **<u>constant background attack</u>**, not occasional spikes, meaning **monitoring must run**

Anomaly Detection

<u>Top 50 highest-risk events</u> <u>based</u> on **extremely high Anomaly Scores.**
Scores near 100 indicate behaviour far outside normal patterns, such as unusual **<u>payload sizes, unexpected port access, or abnormal traffic timing.</u>** These events represent the "**<u>needle in the haystack</u>**" and are the highest priority for forensic investigation.



Automation Strategy

1. Collect – A **Python script calls the API every 5–10 minutes** to fetch only new events.

2. Ingest – The script **converts the data to JSON and sends it into Splunk via HTTP Event Collector (HEC).**

3. Visualise – **Dashboards are configured for Real-tim**e or Last **24 hours** to show the latest incidents.

4. Alert – **Splunk alerts trigger e-mails or tickets** when thresholds are exceeded (e.g. spike in High severity).

This keeps dashboards continuously updated without manual uploads.

Conclusion

Findings:

- Firewall policy failure – **High/Critical attacks were not blocked** as required.
- **Continuous attack trend** – threats appear every month, indicating a persistent landscape.
- High-risk anomalies – a **small set of events** with **extreme anomaly scores** require urgent review.

Recommendations:

- **Fix firewall/IPS rules** so all High/Critical events are blocked and alerted.
- **Implement automated API** → HEC ingestion to keep data flowing into Splunk.
- Use **anomaly scores to prioritise analyst investigation time.**